

Math 242, S13
Quiz 3

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the particular solution of the initial value problem:

$$x \frac{dy}{dx} = 2y + x^3 \sin x \cos x, \quad y(\pi) = 1.$$

(First transform the differential equation into the standard form of linear first-order differential equation if it is not.)

Solution: We transform the above equation into the standard form

$$\frac{dy}{dx} = \frac{2y}{x} + x^2 \sin x \cos x$$

$$\frac{dy}{dx} - \left(\frac{2}{x}\right)y = x^2 \sin x \cos x$$

Integrating factor

$$\rho(x) = e^{\int -\frac{2}{x} dx} = e^{-2 \ln |x|} = \frac{1}{x^2}$$

Thus we have

$$\frac{1}{x^2} \frac{dy}{dx} - \frac{1}{x^2} \left(\frac{2}{x}\right)y = \left(\frac{1}{x^2}\right)x^2 \sin x \cos x$$

$$\frac{1}{x^2} \frac{dy}{dx} - \frac{2}{x^3} y = \sin x \cos x$$

$$\frac{d}{dx} \left[\frac{1}{x^2} y \right] = \sin x \cos x$$

$$\frac{1}{x^2} y = \int \sin x \cos x dx$$

$$\frac{1}{x^2} y = \frac{\sin^2 x}{2} + C$$

$$\Rightarrow y(x) = x^2 \left(\frac{\sin^2 x}{2} + C \right)$$

$$y(\pi) = 1 \Rightarrow \pi^2 \left(\frac{\sin^2 \pi}{2} + C \right) = 1 \Rightarrow \pi^2 C = 1 \Rightarrow C = \frac{1}{\pi^2}$$

Thus, the particular solution is

$$y(x) = x^2 \left(\frac{\sin^2 x}{2} + \frac{1}{\pi^2} \right)$$